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Sampling period, statistical complexity, and chaotic attractors

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ABSTRACT

We analyze the statistical complexity measure vs. entropy plane-representation of sampled chaotic attractors as a function of the sampling period τ and show that, if the Bandt and Pompe procedure is used to assign a probability distribution function (PDF) to the pertinent time series, the statistical complexity measure (SCM) attains a *definite maximum for a specific sampling period* t_M. On the contrary, the usual histogram approach for assigning PDFs to a time series leads to essentially constant SCM values for any sampling period τ. The significance of t_M is further investigated by comparing it with typical times found in the literature for the two main reconstruction processes: the Takens' one in a delay-time embedding, on one hand, and the exact Nyquist–Shannon reconstruction, on the other one. It is shown that t_M is compatible with those times recommended as adequate delay ones in Takens' reconstruction. The reported results correspond to three representative chaotic systems having correlation dimension $2 < D_2 < 3$. One recent experiment confirms the analysis presented here.

1. Introduction

The study of randomness started with Poincaré, but it took almost 80 years for digital electronics to make enough computational resources available so that the investigation on the transition from order to randomness could lead to the fascinating issue of deterministic chaos. One hallmark of chaotic systems is the presence of recognizable patterns in their state space. Pattern-formation plays a fundamental role in the definition of (i) structural complexity by Crutchfield and Young [1] and (ii) in a number of other complexity measures, such as the so-called statistical complexity (SCM), relevant for this paper. For an excellent review on complexity measures see Ref. [2]. The SCM, based on the notion of *disequilibrium* in probability space, was proposed by López Ruiz et al. [3] and improved upon later in several papers [4–6]. The SCM version employed here was advanced in Ref. [5]. In a related vein, the importance of using a "*causal*" Probability Distribution Function (PDF) to analyze the stochasticity-degree of chaotic and pseudo stochastic systems was emphasized in Refs. [6–9].

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Recently, it has been shown that chaotic maps in the causality entropy-complexity plane ($H \times C$-plane) display a typical localization behavior [6]. This behavior is also observed for the case of continuous chaotic dynamical systems when they are represented by means of Lorenz map [6]. Accordingly, one would expect that similar planar-performances would be observed for chaotic sampled attractors However, in opposition to the case of maps where the sampling period is $\tau = 1$, one sees that the corresponding value of τ requires careful consideration.

- If τ is too small, little information gain accrues between successive sampled values (the values in the corresponding time series will be almost linearly dependent), a *redundancy effect* [10] that "moves" the associate planar-localization to regions with $H \to 0$.
- If τ is very large, successive sampled values may became unrelated, an *irrelevance effect* [10]. The planar-localization shifts to $H \to 1$ zones.

In this paper we examine the following working hypothesis regarding planar localization behavior: *chaotic maps' idiosyncratic planar behavior will also be encountered in the case of chaotic sampled attractors provided that the chaotic dynamical system is sampled at a specific time t_M, characteristic of the dynamics*. Such a t_M will be the result of an optimal tradeoff between redundancy and irrelevance that allows for a maximization of the statistical complexity.

1.1. Contextual considerations

It is indeed important to study just how the sampling period τ influences the final results, given that digital instrumentation is widely used in all kinds of experimental set-ups. The most common criterion is given by the sampling theorem by Nyquist [11] and Shannon [12]. It stipulates that the *exact reconstruction* of a continuous signal is possible whenever its Fourier spectrum vanishes for frequencies above a certain $B = f_{\max}$ (the bandwidth). An *infinite* number of regularly sampled values is required and τ must satisfy the inequality $\tau < t_{NS} = 1/(2B)$ where t_{NS} is the Nyquist–Shannon minimum sampling interval. The exact original continuous signal is recovered from the time series by using an *ideal low pass filter*. Note that in actual scenarios the number N of samples is always finite and an exact reconstruction is not possible.

In a different vein, Takens [13] demonstrated that chaotic attractors generated by dynamical systems may be *reconstructed* if one has access to the *infinite* time series of regularly sampled values for just one state variable. Takens' procedure allows one to obtain the main parameters characterizing the chaotic system, like Lyapunov exponents, dimensions, etc. For an impressive web-site containing very useful routines for nonlinear time series analysis see Ref. [14]. The ensuing reconstruction procedure uses "delayed-samples" with t_T as the delay time, to generate a d-dimensional vector (in a d-dimensional embedding space). Note that the time delay t_T must be a multiple of the sampling period τ, since we only possess data gathered at those times. Let us remark that in the context of Takens' theorem the vocable *reconstruction* adopts a different meaning than that assigned by the Nyquist–Shannon theorem.

1.2. Present goals

In this paper we differentiate between a *Takens reconstruction* and a *Nyquist reconstruction*. *We will carefully look at just how the representative point of a chaotic attractor in an entropy-complexity plane* ($H \times C$) *changes with the sampling period*. It will be shown that when the *causal PDF* prescribed by the Bandt and Pompe procedure [15] is used, the ensuing complexity $C = C^{(BP)}$ exhibits a maximum for a given sampling period $\tau = t_M$. However, such a maximum does not appear if the conventional histogram approach is used instead in assigning a PDF to the time series and thus a $C = C^{(hist)}$ is used. We call the later histogram-distribution a *non-causal PDF*. In point of fact, that of causality constitutes a difficult topic (see for instance Ref. [16]). In this paper a *causal PDF* is one that takes into account the temporal correlation between successive samples, which entails using a PDF with a symbol assigned to each trajectory's piece of length $L = d \cdot \tau$.

It is of interest, for both the Takens and the Nyquist–Shannon reconstruction processes, to analyze the relation between (i) t_M and (ii) other related temporal quantities proposed in the literature such as t_T or t_{NS}. This is one of the topics to be investigated here. In the case of Takens' reconstruction it is well known that for a finite number of samples N the quality of the reconstruction requires the selection of an adequate value for t_T. There exist in the literature several different recommendations for a "good" t_T, like those based on Ref. [17]: (a) the linear autocorrelation R (the first zero crossing of R, if exist, denoted by t_0; the first minimum of R, denoted by t_1; the first curvature change of R, denoted by t_2; and, the value for which the decay of R is less than R_0/e, denoted by $t_{1/e}$) and (b) average mutual information I (first minimum, denoted by t_I). For an interesting discussion see Ref. [18,19]. In the case of Nyquist's reconstruction, a bandwidth criterion must be chosen in order to evaluate the Nyquist–Shannon time. We denoted it by $t_{NS}^{(\alpha)}$, with $0.80 \leq \alpha \leq 0.99$ the proportion of the power spectrum to be considered due to the spectrum of chaotic systems is not band-limited.

In Section 2 the specific SCM and the PDF used in the paper are reviewed. Our main results concern three paradigmatic examples: (a) the Lorenz Chaotic Attractor; (b) the Rössler Chaotic Attractor and (c) the chaotic attractor named $\mathbf{B}_7$ by Chlouverakis and Sprott [20]. They are reported in Section 3. The first two attractors have been extensively studied in the literature but they both have a correlation dimension $D_2 \approx 2$ while the attractor $\mathbf{B}_7$ has $D_2 \approx 2.719$, covering almost all the 3D state space. Conclusions and remarks are included in Section 4. It is interesting to remark that the analysis performed here has been experimentally verified [21].

2. Tools of the trade

2.1. Statistical complexity measure

The SCM, denoted by C, is an informational quantifier. As such, it is a functional of a PDF, here associated to a *time series*. Given the PDF, $P \equiv \{p_i; i = 1, \ldots, N\}$, there are several manners to obtain an appropriate functional. For a comparison amongst different complexity measures see the excellent paper by Wackerbauer et al. [2]. We adopt the functional form introduced in López Ruiz–Mancini–Calbet seminal paper [3] with the modifications advanced by Lamberti et al. [5], given by

$$C[P] = Q_J[P, P_e] \cdot H[P], \tag{1}$$

where H denotes the amount of "disorder" given by the normalized Shannon entropy $H[P] = S[P]/S_{\max}$ (with $S_{\max} = S[P_e] = \ln N$) and Q_J is the so-called "disequilibrium", defined in terms of the extensive Jensen–Shannon divergence (which in turn induces a squared metric) [5]: $Q_J[P, P_e] = Q_0 \cdot J[P, P_e] = Q_0 \cdot \{S[(P + P_e)/2] - S[P]/2 - S[P_e]/2\}$. In these equations $P_e = \{1/N, \ldots, 1/N\}$ stands for the *uniform* distribution, while $S[P] = -\sum_{j=1}^{N} p_j \ln(p_j)$ is the Shannon entropy corresponding to the PDF P and Q_0 is a normalization constant. We have then $0 \le H \le 1$ and $0 \le Q_J \le 1$.

The complexity measure constructed in this way is intensive, as many thermodynamic quantities [5]. We stress the fact that the above SCM is not a trivial function of the entropy because it depends on two different probability distributions, the one associated to the system under analysis, P, and the uniform distribution, P_e. Furthermore, it was shown that for a given H value, there exists a range of possible SCM values [22]. Thus, it is clear that C carries important additional information (related to the correlational structure between the components of the physical system) that is not contained in the entropic functional.

A detailed analysis of the C-behavior demonstrates the existence of bounds to C that we called $C_{\max}$ and $C_{\min}$. These bounds can be systematically evaluated by recourse to a careful geometric analysis performed in the space of probabilities Ω [22]. The corresponding values of $C_{\max}$ and $C_{\min}$ depend only on the probability space's dimension and, of course, on the functional form adopted by the amount of disorder H and the disequilibrium Q. $H \times C$ diagrams are important and yield system's information independently of the values that the different control parameters may adopt. The bounds also provide us with relevant information that depends on the system's particular characteristics, as for instance, the existence of global extrema, or the peculiarities of the system's configuration for which such extrema obtain.

2.2. The probability distribution function

The PDF P itself is not a uniquely defined object and several approaches have been employed in the literature to "associate" P to a given time series. Just to mention some frequently used P-extraction (from the time series) procedures, one has: (a) frequency counting [23], (b) time series histograms [8], (c) binary symbolic dynamics [24], (d) Fourier analysis [25], (e) wavelet transforms [26,27], (f) permutation (Bandt–Pompe) entropies [15,28], among others. There is ample liberty to choose among them and the specific application must be carefully analyzed so as to make a good choice. Rosso et al. [6] showed that the last mentioned methodology may be profitably used in the plane $H \times C$ so as to separate and differentiate amongst stochastic, chaotic, and deterministic systems. It was shown in Refs. [29,30] that temporal correlations are nicely displayed by the Bandt and Pompe PDF.

Thus, different *symbolic sequences* may be assigned to a given time series. If one symbol a of the finite alphabet $\mathfrak{A}$ is assigned to each x_t of the time series, the *symbolic sequence* can be regarded as a *non causal coarse grained* description of the time series because the resulting PDF will not have any *causal information*. The usual histogram technique corresponds to this kind of assignment. For extracting P via a histogram one divides the interval $[a, b]$ (with a and b the minimum and maximum values in the time series) into a finite number N_{bin} of non overlapping equal sized consecutive subintervals $A_i : [a, b] = \bigcup_{i=1}^{N_{bin}} A_i$ and $A_i \bigcap A_j = \emptyset \,\forall i \neq j$. Of course, in this approach the temporal order of the time-series plays no role at all. The quantifiers obtained via the ensuing PDF-histogram are called in this paper, respectively, $H^{(hist)}$ and $C^{(hist)}$. Let us also point out that for time series with a finite alphabet it is relevant to consider a judiciously chosen optimal value for $N_{bin} \equiv N$ (see e.g. Ref. [8]).

Causal information may be duly incorporated into the construction process that yields P if one symbol of a finite alphabet $\mathfrak{A}$ is assigned to a trajectory's portion, i.e., we assign "words" to each trajectory portion. The Bandt and Pompe methodology for extraction of the PDF corresponds to this type of assignment and the resulting probability distribution P is thus a *causal coarse grained* description of the system. Note that, in the Bandt and Pompe approach a sort of *coarse graining* and *word construction* is effected. Although other ways of get a causal PDF exist, the advantage of the Bandt and Pompe approach lies in the fact that it solves the problem of finding generating partitions. Thus one expect that for increasing patterns' length (embedding dimension) the Bandt and Pompe approach will retain all relevant essentials of the original continuous (in space) dynamics. The quantifiers obtained by appeal to this PDF are denoted in this paper as $H^{(BP)}$ and $C^{(BP)}$, respectively. A notable Bandt and Pompe result consists in yielding a clear improvement on the quality of Information Theory-based quantifiers [6–9,29–43]. For details of the Bandt and Pome approach the reader is referred to Refs. [15,40].

The probability distribution P is obtained once we fix the embedding dimension d and the embedding delay T. The former parameter plays an important role for the evaluation of the appropriate probability distribution, since d determines the number of accessible states, given by $d!$. Moreover, it was established that the length M of the time series must satisfy the condition $M \gg d!$ in order to achieve a proper differentiation between stochastic and deterministic dynamics [6]. With respect to the selection of the parameters, Bandt and Pompe suggest in their cornerstone paper [15] to work with $3 \leq d \leq 7$ with a time lag $T = 1$. This is what we do here (in the present work $d = 6$ and $T = 1$ are used). Of course it is also assumed that enough data are available for a correct attractor reconstruction.

3. Results

3.1. Preliminaries

Our results refer to three paradigmatic chaotic systems with a 3-dimensional state space, namely,

- The Rossler chaotic attractor, given by

$$\begin{cases} \dot{x} = b + x\,(y - c) \\ \dot{y} = -x - z \\ \dot{z} = y + a\,z, \end{cases} \tag{2}$$

where the parameters used here are $a = 0.45$, $b = 2$, and $c = 4$, corresponding to a chaotic attractor.

- The Lorenz chaotic attractor given by

$$\begin{cases} \dot{x} = \sigma\,(y - x) \\ \dot{y} = rx - y - xz \\ \dot{z} = xy - bz, \end{cases} \tag{3}$$

where the pertinent parameters are $\sigma = 16$, $b = 4$, and $r = 45.92$, corresponding to a chaotic attractor.

- The chaotic attractor $\mathbf{B}_7$ of Ref. [20]:

$$\begin{cases} \dot{x} = K + z\,(x - \alpha\,y) \\ \dot{y} = z\,(\alpha\,x - \epsilon\,y) \\ \dot{z} = 1 - x^1 - y^2, \end{cases} \tag{4}$$

where $K = 0.5$, $\alpha = 7.0$ and $\epsilon = 0.23$.

The first two systems exhibit strong differences that make them interesting for this study. The Rossler oscillator "spends" most of the time near the plane y, z with $x \simeq 0$. The time series obtained by sampling the x coordinate looks like a *delta train* as Fig. 1a illustrates. The Power Spectrum (PS) displayed in Fig. 2a shows that the characteristic periodicities are very similar for all the three coordinates, with a higher frequency-content for x, produced by its delta like shape. On the other hand, the Lorenz oscillator displays a very different spectrum for coordinate z (see Fig. 2b), as compared with the PS of x or y. The reason is that the attractor spirals from the center to the border of one of the typical butterfly's wings and then suddenly moves to the other wing, as Fig. 1b shows. Both attractors have correlation dimension D_2 close to 2. The attractor $\mathbf{B}_7$ has a $D_2 = 2.719 \pm 0.156$ [20], covering most of the available state-space. In Figs. 1c and 2c the time evolution of the three coordinates and their corresponding PS are displayed.

The evolution of each dynamical system was determined by recourse to a variable-step Runge–Kutta–Fehlberg approach [44]. Evaluations were made using sampling periods τ ranging from (i) 0.01 to 5, in steps $\Delta\tau = 0.01$ for the Rossler system, (ii) 0.001 to 0.3 in steps of $\Delta\tau = 0.001$ for the Lorenz system, and (iii) 0.001 to 1 in steps of $\Delta\tau = 0.001$ for $\mathbf{B}_7$ system. With these τ's one covers all the interesting regions: (1) the over-sampled one where τ is very small as compared with the characteristic times considered, and (2) the under-sampled region where τ is very large as compared with the characteristic times. For every τ-value, 10 realizations were generated by starting from different initial conditions. The time series for the state variables x, y, and z were stored in a matrix of 3×10^5, after skipping the first 10^4 iterations in order that transient states die out.

We employed Bandt and Pompe's procedure (with dimension $d = 6$ and time lag $T = 1$) to assign a PDF to each time series. These PDF's exhibit an entropy $H^{(BP)}$ and a complexity $C^{(BP)}$. Functionals $H^{(hist)}$ and $C^{(hist)}$ were evaluated as well, with a PDF obtained from the pertinent histogram. In this case, the range covered by each variable was divided into 2^{16} uniformly distributed subintervals. The mean values $\langle C \rangle$ and $\langle H \rangle$ were computed for all the here considered PDF-functionals and all the realizations. For simplicity, the symbol $\langle \bullet \rangle$, meaning mean value over realizations, is suppressed.

3.2. Our findings

We start presenting our results at this stage. The entropy-complexity plane $H^{(BP)} \times C^{(BP)}$ is quite useful for distinguishing between stochastic noise and deterministic chaotic behavior [6]. In such a vein we first analyze the planar evolution of representative points for our systems as the sampling frequency changes (see Fig. 3a–c for the three sampled variables of the Rossler system, Fig. 3d–f for the Lorenz system, and Fig. 3g–i for the $\mathbf{B}_7$ system). As was explained in Section 2.1, "limiting curves" in the plots Fig. 3 determine the allowed values of entropy and complexity. There exists no PDF with entropy-complexity coordinates outside the region limited by these curves [22].

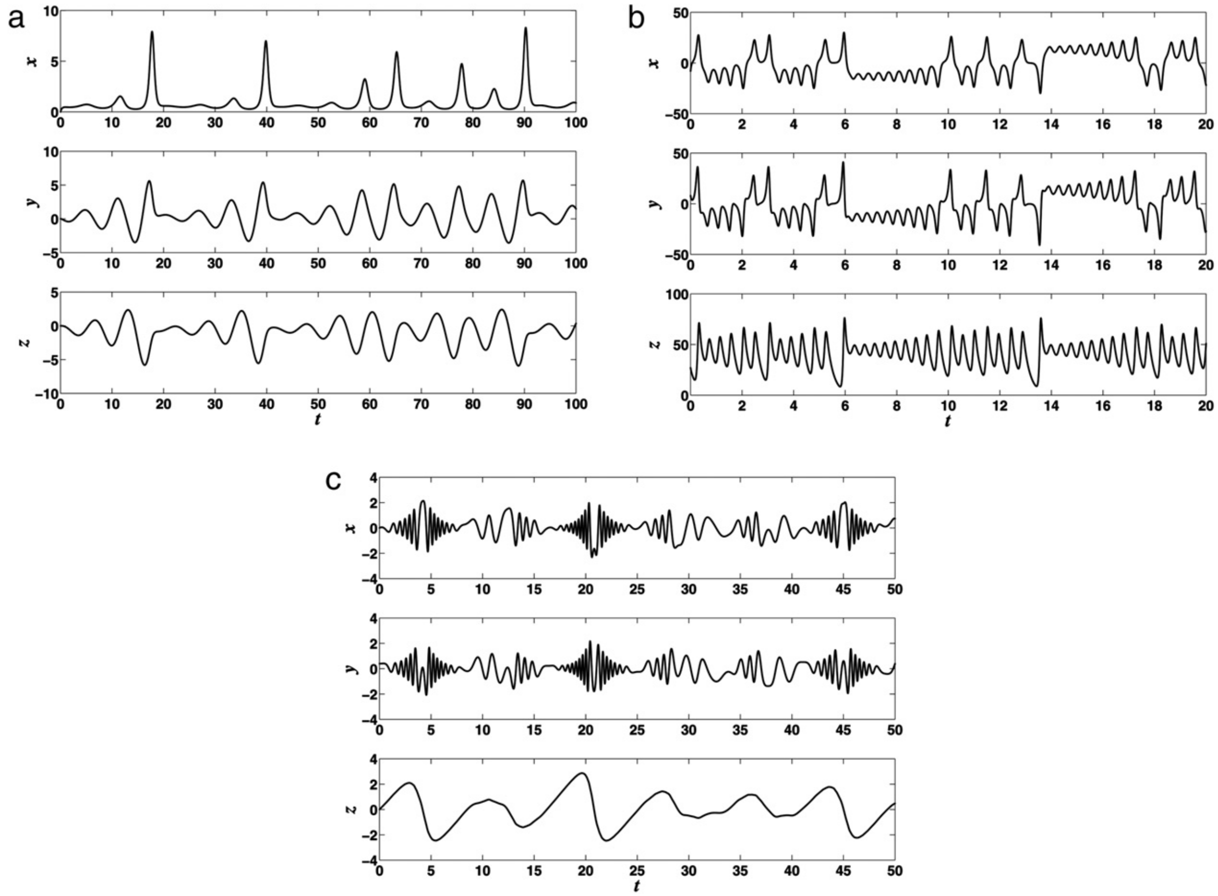

Fig. 1. Evolution of the three coordinates as functions of time: (a) Rossler variables x, y, and z for $\tau = 0.01$ with $a = 0.45$, $b = 2$, and $c = 4$. (b) Lorenz variables x, y, and z for $\tau = 0.001$ with $\sigma = 16$, $r = 45.92$, and $b = 4$. (c) $\mathbf{B}_7$ variables x, y, and z for $\tau = 0.001$ with $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$.

As the sample period τ decreases, the representative point evolves in the $H^{(BP)} \times C^{(BP)}$ plane, from the right planar region with low complexity and high entropy (under-sampling system with small sampling frequency $f = 1/\tau$) to the left region with low complexity and low entropy (over-sampling with big sampling frequency $f = 1/\tau$). The reason for such behavior is, as we comment in Section 1, that for a very big sampling period the dynamical correlations between consecutive measures are lost due to the mixing behavior of the chaotic system under study. That is, the measures can be completely uncorrelated and present characteristics like random behavior: $H \approx 1$ and $C \approx 0$. On the contrary, for very small sampling periods τ the measures will be (strongly) time-correlated so that successive measures suggest being in the presence of a very ordered system: $H \approx 0$ and $C \approx 0$. The evolution of the system in the $H^{(BP)} \times C^{(BP)}$ plane for regimes intermediate between these two extreme sampling-period instances will depend on the specific characteristics of the nonlinear system under study: whether it presents mixing or not; its chaos-degree, etc.

It is important to keep in mind that non linear correlations are the origin of the geometric structures that constitute the hallmark of deterministic chaos [6]. Precisely, in previous works $C^{(BP)}$ was seen to be a measure of the complexity induced by these non linear correlations. Our present results tell us that $f_M = 1/t_M$ – the sampling frequency corresponding to the point with maximum complexity – can be regarded as the *optimum sampling frequency*, i.e., the minimum sampling frequency that retains all the information concerning these structures. Higher sampling frequencies imply an oversampling (*redundancy effect*), producing unnecessary long files to cover the full attractor, while lower sampling frequencies make the time series to lose some vital information concerning nonlinear correlations (*irrelevant effect*). In the over-sampled case, specific d-length ordinal patterns $(0, 1, \ldots, d-1)$ and $(d-1, \ldots, 1, 0)$ appear more frequently than any other ordinal pattern, causing a low entropy $H^{(BP)}$ and a low complexity $C^{(BP)}$. Points in the region with high entropy and low complexity of the $H^{(BP)} \times C^{(BP)}$ plane correspond to under-sampled trajectories of the continuous system. The samples are not correlated at all and the system behaves randomly. Consequently, all the ordinal patterns appear with almost the same frequency, as the ensuing high entropies $H^{(BP)}$ and low complexities $C^{(BP)}$ reveal.

Let us stress the importance of using the Bandt and Pompe procedure to obtain the all-important PDF. A well-known result is that any chaotic map shares with its iterated maps the same histogram [45]. This property is the basis of the *skipping randomizing technique* [8,46]. For that reason De Micco et al. [8] proposed the use of two different PDFs, as relevant for testing

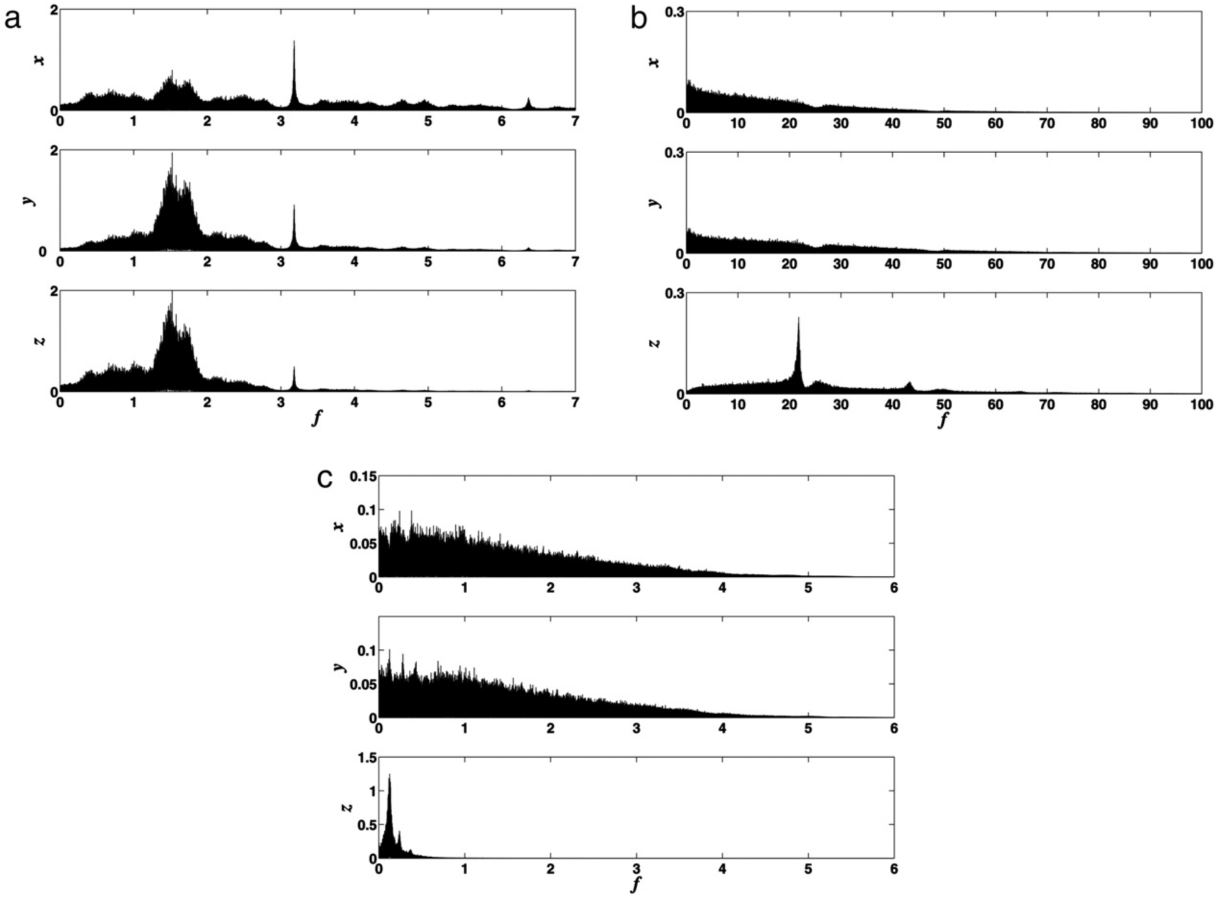

Fig. 2. Power spectra of the coordinates x, y, and z as a function of frequency: (a) Rossler system with $\tau = 0.01$ with $a = 0.45$, $b = 2$, and $c = 4$. for $\tau = 0.01$, and $\Delta\tau = 0.01$. (b) Lorenz system with $\sigma = 16$, $r = 45.92$, and $b = 4$ for $\tau = 0.001$, and $\Delta\tau = 0.001$. (c) $\mathbf{B}_7$ system with $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$ for $\tau = 0.001$, and $\Delta\tau = 0.01$. In all the cases the time series length considered had $M = 10^5$ *data*.

(i) the uniformity of $\mu(x)$ (the invariant measure) and (ii) the mixing constant r_{mix} pertaining to a chaotic time series, as a means to generate pseudo random numbers. They proceeded to employ (a) a PDF P_1 based on time series' histograms and (b) another PDF P_2 based on ordinal patterns (permutation ordering) that derives from appealing to the already cited Bandt and Pompe method [15]. One may conjecture that continuous systems exhibit an analogous property, viz., the sampling process does not significantly change the histogram of the time series and, consequently, any quantifier based on the PDF histogram is not sensitive to the sampling frequency.

Such a conjecture is validated in Fig. 4, where the statistical complexities $C^{(BP)}$ and $C^{(hist)}$ are depicted as functions of the sampling frequency for the x-coordinates of the three systems (see Fig. 4a for Rossler's instance, Fig. 4b for Lorenz' case, and Fig. 4c for $\mathbf{B}_7$). A similar result is obtained if $C^{(BP)}$ and $C^{(hist)}$ are evaluated for the other coordinates y and z, and also if mean values $C^{(BP)}_{(xyz)} = [C^{(BP)}_{(x)} + C^{(BP)}_{(y)} + C^{(BP)}_{(z)}]/3$ and $C^{(hist)}_{(xyz)} = [C^{(hist)}_{(x)} + C^{(hist)}_{(y)} + C^{(hist)}_{(z)}]/3$ are used. Consequently, we conclude that the usual histogram technique is not useful to analyze the effects of the sampling frequency and that a causal PDF (here the Bandt and Pompe one) *must* be employed instead. Note that in Figs. 3 and 4 many irregularities of the trajectories become visible. These are produced by the nontrivial relation between $C^{(BP)}$ and $H^{(BP)}$, and diminish as the number of realizations increases. Note that in Fig. 4, where $C^{(BP)}$ is represented as a function of $f = 1/\tau$, a maximum is easily detected.

Table 1 summarizes our results. $t_M^{(x)}$, $t_M^{(y)}$, and $t_M^{(z)}$ are the inverses of the *sampling frequency* for each coordinate producing the maximum $C^{(BP)}$. We have evaluated $t_M^{(x)}$, $t_M^{(y)}$, and $t_M^{(z)}$ for each realization and then determined the associated mean values and standard deviations shown in columns 1–6. The mean value of $C^{(BP)}$ over coordinates x, y, and z also exhibits a maximum for the t_M indicated in column 7 ($\langle x, y, z\rangle$). All the characteristic times used are displayed in Table 1. Column 8 displays the ratios between each characteristic time and the t_M of column 7. Figures closer to unity in column 8 (within each class of characteristic times) are emphasized by using bold-types.

The normalized Shannon entropy and statistical complexity evaluated with PDF-Bandt and Pompe are not system invariant (neither are other quantifiers derived by Information Theory). The values of these quantifiers depend on how the PDF is evaluated. However, they provide very useful information about the dynamics of the system under study. It is

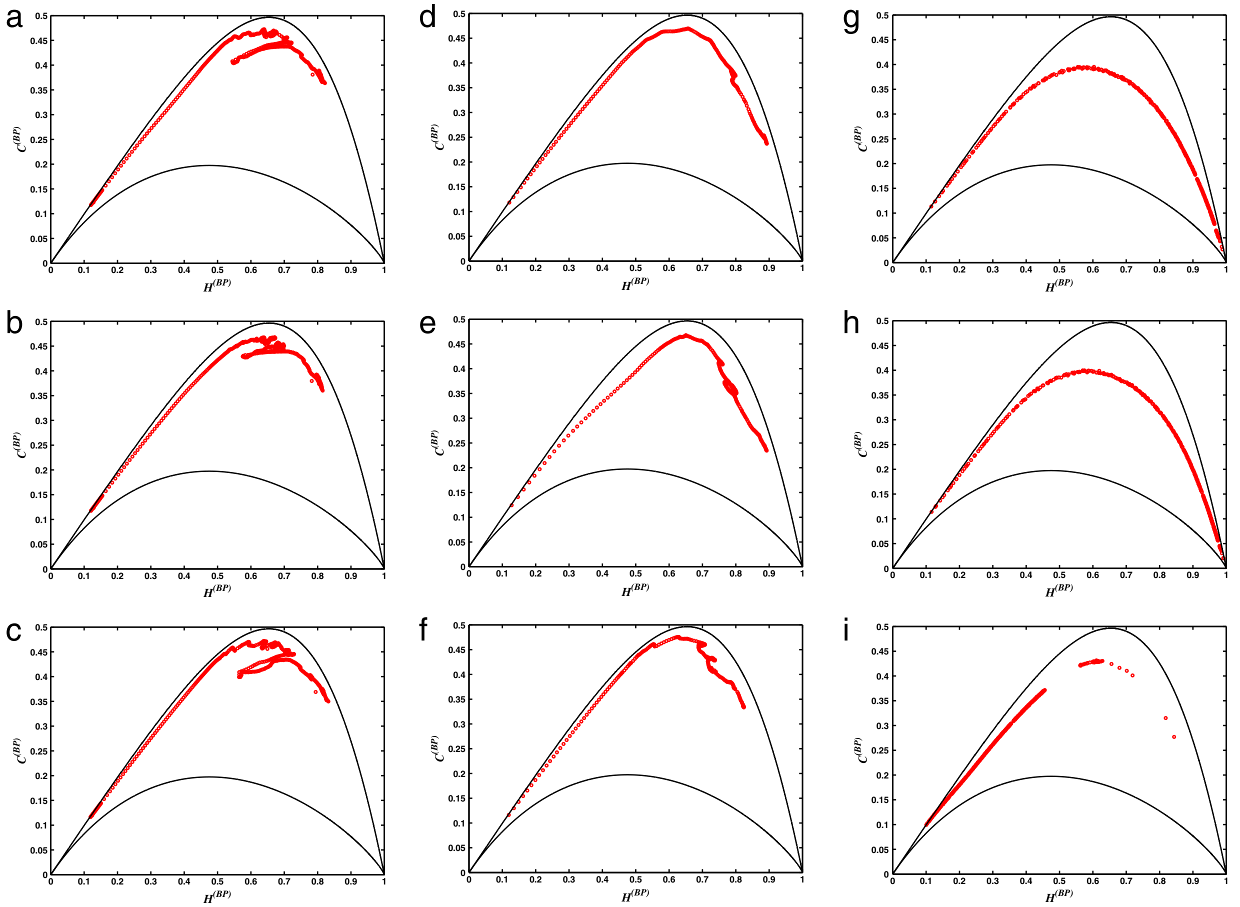

Fig. 3. Representation on the entropy-complexity plane: (a–c) Rossler variables x, y, and z for $0.01 \leq \tau \leq 5$, $\Delta\tau = 0.01$, with $a = 0.45$, $b = 2$, $c = 4$. (d–f) Lorenz variables x, y, and z for $0.001 \leq \tau \leq 0.3$, $\Delta\tau = 0.001$, with $\sigma = 16$, $r = 45.92$, and $b = 4$. (g–i) $\mathbf{B}_7$ variables x, y, and z for $0.001 \leq \tau \leq 3$, $\Delta\tau = 0.001$, with $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$. Under-sampling corresponds to the high $H^{(BP)}$ and low $C^{(BP)}$ region, and over-sampling to low $H^{(BP)}$ and low $C^{(BP)}$. In all the cases the time series length considered had $M = 10^5$ *data*. The continuous lines represent, respectively, the maximum and minimum complexity values (for $d = 6$) for a fixed value of the entropy.

clear then that $H^{(BP)}$ and $C^{(BP)}$ and the quantities derived from them could attain different values according with the specific time series considered (i.e. time series generated by the different coordinates of the system or a combination of them) even if it is reasonable to expect that their corresponding values do not exhibit a strong dispersion. This assumption is clearly confirmed if we look at the values obtained for t_M in Table 1, evaluated for different time series. As expected, the same kind of behavior is obtained in the case of t_I (Average Mutual Information).

From inspection of the results displayed in Table 1 we see that the time t_M yields similar values to those obtained from the evaluation of the Average Mutual Information t_I. The largest difference between t_M and t_I (see column 8 in Table 1) is observed for the $\mathbf{B}_7$ system. Such disparity can be attributed to the fact that this system possesses an attractor with $D_2 \approx 3$ that covers most of the state-space in contrast with the other two systems considered here, the Lorenz and Rossler ones, which have $D_2 \approx 2$ [19,47] (see Fig. 5).

With regards to t_M and the others special characteristic times (for our three dynamical systems) we can state that, in the Rossler-case, such characteristic times are, respectively, the first minimum of the mutual information t_I, the first zero of the autocorrelation t_0, and t_{NS}, for $\alpha = 0.85$. In the the Lorenz instance the characteristic times closest to t_M are: the first minimum of the mutual information t_I, the first change of curvature of the autocorrelation t_2, and t_{NS} for $\alpha = 0.9$. Finally, for $\mathbf{B}_7$ the characteristic times closest to t_M are: $t_{1/e}$, and t_{NS} for $\alpha = 0.85$. The first minimum of the mutual information appears far from the t_M-location. The shape of I (discrete mutual information) is very irregular, as shown in Fig. 6c, which explains why such features takes place.

For all the systems dealt with here, the Nyquist criterion provides smaller characteristic times if $\alpha \geq 0.95$ is used instead, illustrating thus the difference between the *exact reconstruction* that is the goal of the Nyquist–Shannon Theorem and the *approximate attractor reconstruction* that relies on Takens' theorem. Note that the autocorrelation function is based just on linear statistics and does not contain any information concerning nonlinear dynamical correlations. Kantz and Schreiber [19] stressed the drawback of using linear statistics for the Lorenz system, with a typical trajectory staying some time on one

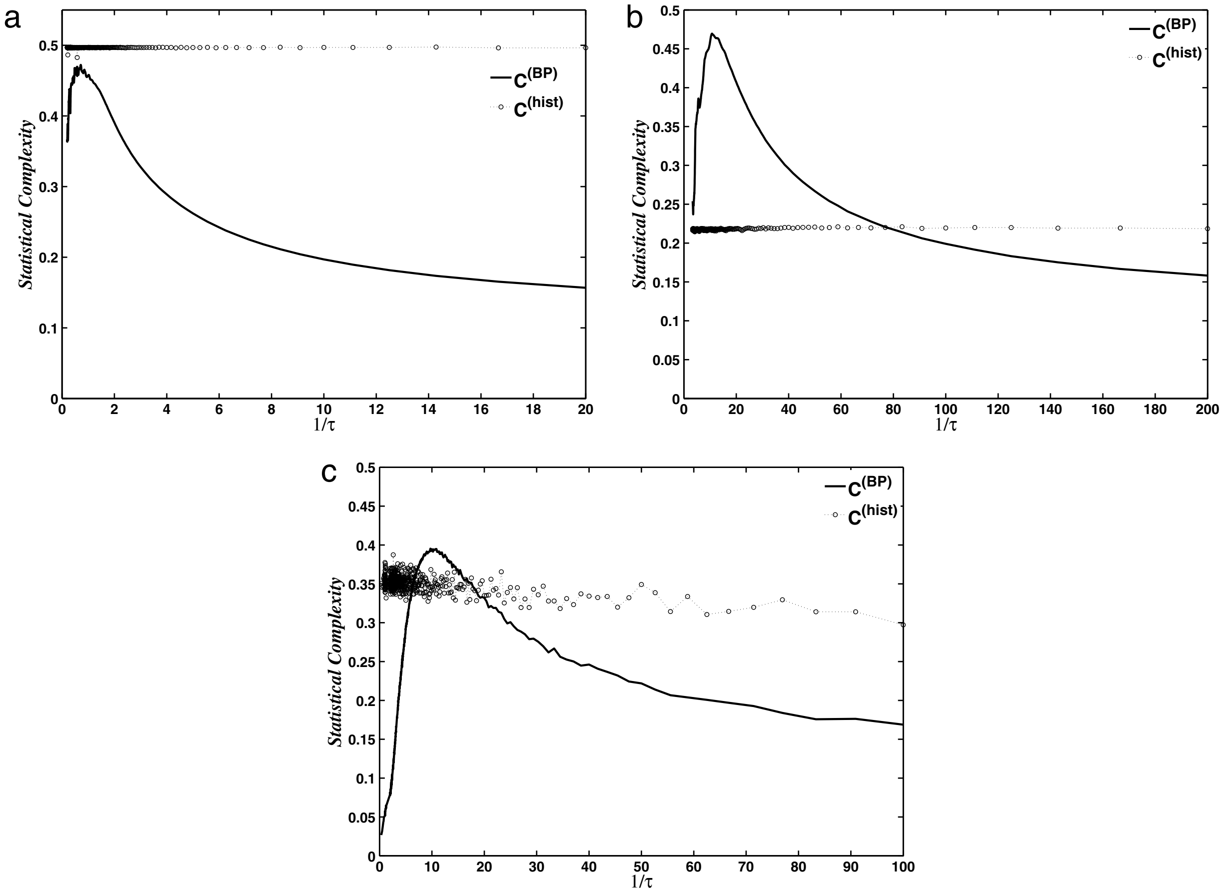

Fig. 4. Intensive statistical complexity measures $C^{(BP)}$ and $C^{(hist)}$ for the coordinate x of the three systems. (a) Rossler system, (b) Lorenz system and (c) $\mathbf{B}_7$ system one. $C^{(BP)}$ is evaluated by using the PDF obtained with the Bandt and Pompe prescription with $d = 6$. $C^{(hist)}$ is evaluated by using the PDF obtained from the histogram of x, using 2^{16} bins. $C^{(hist)}$ is almost constant for all values of τ. In all the cases the time series length considered had $M = 10^5$ *data*. The use of the BP prescription is of the essence in getting the complexity-maxima. The reason is that a change in the sampling frequency does not change the invariant measure of each variable of the chaotic attractor. The situation is identical to that of the skipping procedure in pseudo random number generators based on chaotic maps [8].

wing of the butterfly attractor and spiraling then from the inside to the border before jumping to the other wing. The autocorrelation does not reveal the importance of the period of such "jumping" between wings, but the mutual information does so. Also, the autocorrelation of the squared signal is able to detect this periodic change [19].

So as to assess the quality of the Takens' reconstruction process using t_M, D_2 is evaluated for an embedding dimension ranging from $d_e = 1$ to 10 (see Fig. 7). The pertinent estimation of D_2 is compared with values reported in the literature for the continuous system [19,48,49]. The subroutine *d2* of the TISEAN package [14] was used for calculating D_2. Let us stress that the estimation of D_2 poses a very delicate problem, so that we refer the readers to the documentation available for the TISEAN package and also to the excellent book by Kantz and Schreiber [19]. Figs. 7 show that $\tau = t_M$ produces a time series with a D_2 that turns out to be a better estimation for the correlation dimension of the continuous system than those obtained with higher (under-sampled) or lower (over-sampled) values for τ. Similar results were obtained when we consider the evaluation of the maximum Lyapunov exponent.

Remark that the time t_M, for which one detects a maximum in the entropy-complexity plane $H^{(BP)} \times C^{(BP)}$, is unique for the three dynamical systems analyzed in the present work. We associate this sample time with the one needed to capture the most relevant insights related to the dynamics' nonlinear correlations.

Recently Soriano et al. [21] theoretically and experimentally studied a chaotic semiconductor laser with optical feedback. This is an extremely interesting system because dynamical systems with time delay, like the Mackey–Glass or a laser with delay feedback, exhibit different relevant time scales [21,43]. Consequently, additional complexity-maxima could be expected (in addition to the one encountered in dealing with the chaotic systems of this paper). In the case of the laser studied in Ref. [21] the main time scales are: (i) τ_S^*, the feedback-time providing the largest time scale, (ii) T_{RO}, that is the relaxation oscillation providing a shorter time scale, and finally (iii) chaotic oscillations, that govern the fastest time scale. Soriano et al. experimentally confirmed that our prescription for t_M corresponds to a maximum of $C^{(BP)}$, as found in all the

Table 1
Characteristic times for coordinates x, y, z and their standard deviation (denoted by σ_i) of the Rossler, Lorenz and $\mathbf{B}_7$ chaotic attractors. The system's parameters are $a = 0.45$, $b = 2$, $c = 4$ for Rossler, $\sigma = 16$, $r = 45.92$, and $b = 4$ for Lorenz and $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$ for $\mathbf{B}_7$. In all the cases the time series length considered had $M = 10^5$ *data*. $\langle x, y, z\rangle$ represent the corresponding time obtained averaging $C^{(BP)}$ over the three coordinates. Each characteristic time is classified according with the method used to obtain it. Numbers in bold face indicate the characteristic time of each method that best approximates t_M (the time corresponding to the maximum of the statistical complexity measure $C^{(BP)}$) ($d = 6$).

Time	x	σ_x	y	σ_y	z	σ_z	$\langle x, y, z\rangle$	t/t_M
Rossler system								
Time induced by complexity								
t_M	1.3970	0.0067	1.8090	0.1473	1.4050	0.0053	1.4000	1.000
Times induced by autocorrelation								
t_0	1.4560	0.0178	1.5600	<.00001	1.7150	0.0118	1.6000	**1.1429**
t_1	3.2070	0.0157	3.0820	0.0063	3.2350	0.0108	3.1600	2.2571
t_2	0.6380	0.0079	1.3520	0.0103	1.5190	0.0032	1.0000	0.7143
$t_{1/e}$	0.8480	0.0157	1.1700	0.0063	1.2930	0.0108	1.1200	0.8000
Time induced by mutual information								
t_I	1.6240	0.0534	1.4530	0.0874	1.5600	0.0337	1.5200	**1.0857**
Times induced by Nyquist–Shannon								
$t_{NS}^{(0.85)}$	0.8795	–	1.4645	–	2.0956	–	1.4799	**1.0571**
$t_{NS}^{(0.90)}$	0.7292	–	1.1418	–	1.6661	–	1.1791	0.8422
$t_{NS}^{(0.95)}$	0.5691	–	0.7447	–	0.9142	–	0.7427	0.5305
Lorenz system								
Time induced by complexity								
t_M	0.0932	0.0004	0.0722	0.0010	0.0853	0.0007	0.0860	1.0000
Times induced by autocorrelation								
t_0	2.3805	1.0482	1.9424	1.1457	0.1217	0.0018	0.2130	2.4767
t_1	0.4226	0.0308	0.3979	0.0342	0.2334	0.0007	0.2580	3.0000
t_2	0.1012	0.0006	0.0760	0.0001	0.0874	0.0005	0.0860	**1.000**
$t_{1/e}$	0.2078	0.0308	0.1478	0.0342	0.0883	0.0007	0.1230	14.3023
Time induced by average mutual information								
t_I	0.1048	0.0006	0.0970	0.0012	0.0941	0.0028	0.0990	**1.1512**
Times induced by Nyquist–Shannon								
$t_{NS}^{(0.85)}$	0.1383	–	0.1082	–	0.0884	–	0.1116	1.2981
$t_{NS}^{(0.90)}$	0.1138	–	0.0880	–	0.0670	–	0.0896	**1.0419**
$t_{NS}^{(0.95)}$	0.0756	–	0.0639	–	0.0084	–	0.0493	0.5733
$\mathbf{B}_7$ system								
Time induced by complexity								
t_M	0.1030	<0.001	0.0930	<0.001	0.9540	<0.001	0.3833	1.0000
Times induced by autocorrelation								
t_0	0.5459	0.3775	0.5635	0.4363	1.9985	0.0258	1.9860	5.1808
t_1	0.4509	0.1332	0.3915	0.0957	3.9141	0.1161	3.6870	9.6182
t_2	0.1301	0.0129	0.1301	0.0127	1.5631	0.0227	0.1280	0.3339
$t_{1/e}$	0.1813	0.1332	0.1747	0.0957	1.4504	0.1161	0.2830	**0.7382**
Time induced by average mutual information								
t_I	1.5289	0.2240	1.5951	0.2212	1.7905	0.1041	1.6551	**4.3176**
Times induced by Nyquist–Shannon								
$t_{NS}^{(0.85)}$	0.1901	–	0.1968	–	0.5319	–	0.3062	**0.7987**
$t_{NS}^{(0.90)}$	0.1683	–	0.1736	–	0.0731	–	0.1383	0.3607
$t_{NS}^{(0.95)}$	0.0539	–	0.1441	–	0.0076	–	0.06853	0.1787

chaotic systems studied in this paper (see Fig. 10 in Ref. [21]). To our knowledge, this is the first controlled experiment where our prescription has been to hold.

Sharing the point of view of Abarbanel [47], our prescription for the sample period $\tau = t_M$ is based on the consideration of a fundamental aspect of chaos, namely, the generation of information. Thus, our choice is made by taking into account an important property of the system we wish to describe. Remark that stable linear systems generate zero information. In consequence, information generation [47] is a property of nonlinear dynamics not shared by linear time evolutions.

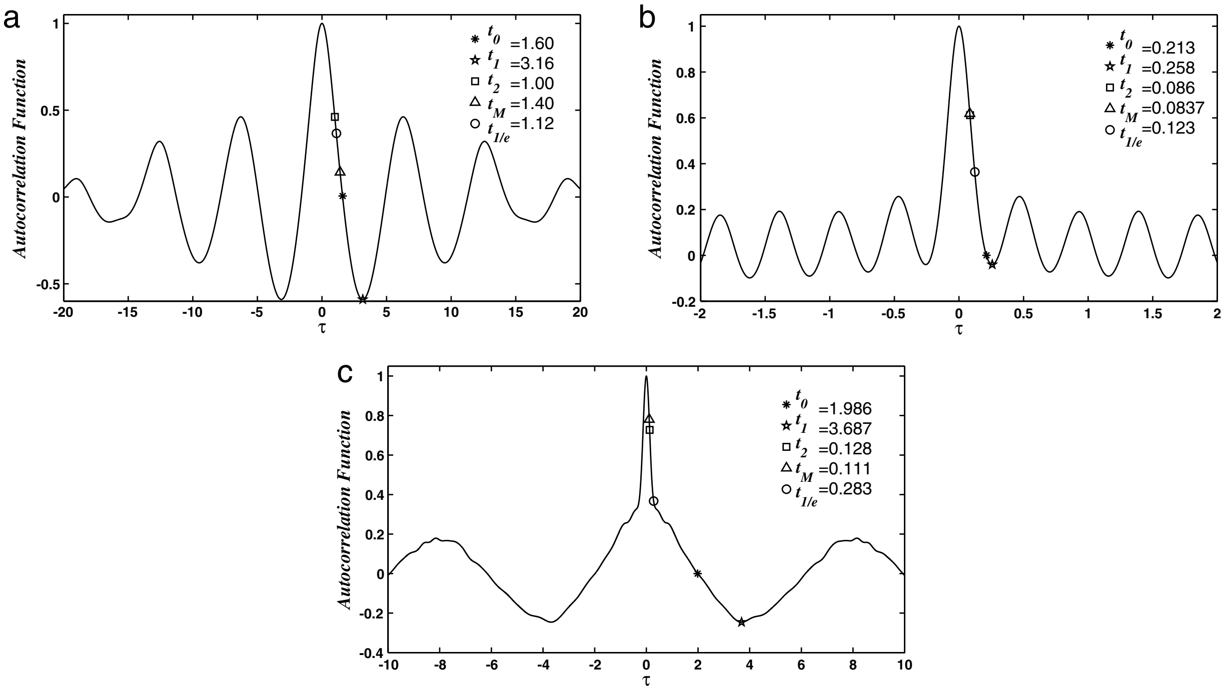

Fig. 5. Discrete autocorrelation function R_m for: (a) Rossler system with $a = 0.45$, $b = 2$, $c = 4$. (b) Lorenz system with $\sigma = 16$, $r = 45.92$, and $b = 4$. (c) $\mathbf{B}_7$ system with $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$. In all the cases the time series length considered had $M = 10^5$ *data*. We show both the characteristic time induced by $C^{(BP)}$ ($d = 6$), that is t_M, and its most approximate characteristic time induced by R_m.

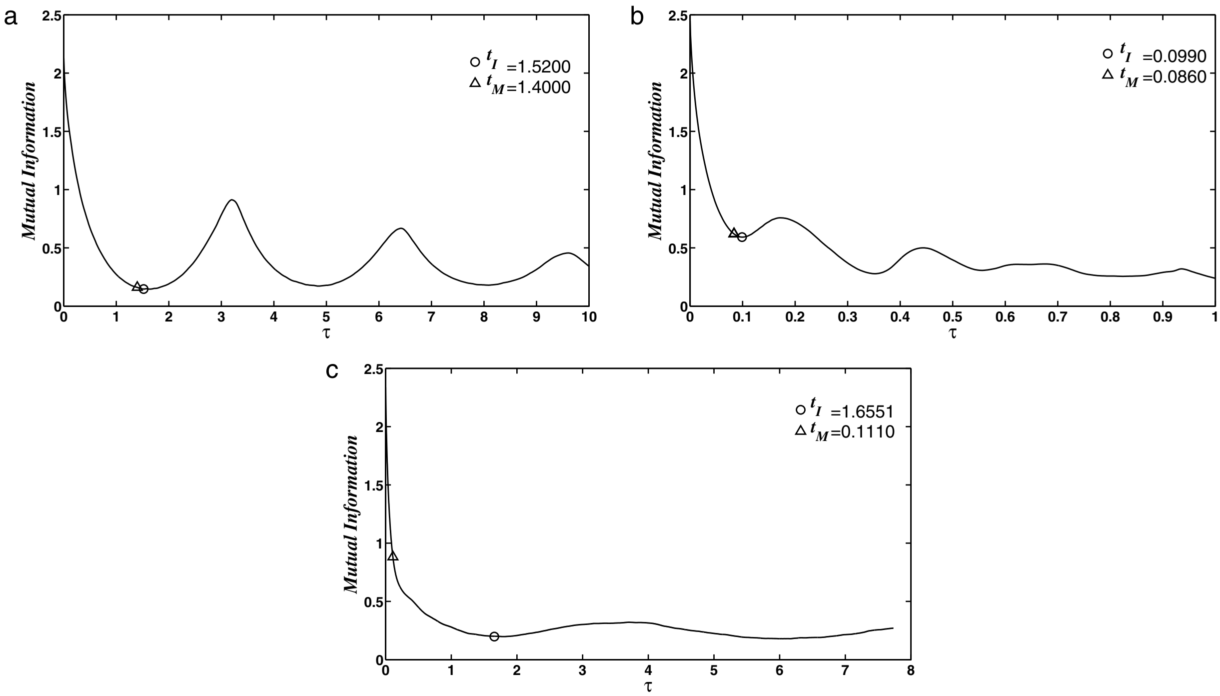

Fig. 6. Discrete mutual information I for: (a) Rossler system with $a = 0.45$, $b = 2$, $c = 4$. (b) Lorenz system with $\sigma = 16$, $r = 45.92$, and $b = 4$. (c) $\mathbf{B}_7$ system with $K = 0.5$, $\alpha = 7.0$ and $\epsilon = 0.23$. In all the cases the time series length considered had $M = 10^5$ *data*. The characteristic time induced by $C^{(BP)}$ ($d = 6$), that is t_M and its closest-lying characteristic time induced by I_m are shown.

Finally, another issue to be aware of is concerns the influence of noise. In order to verify the robustness of the criterion here advanced, Gaussian noise was added to each system and the ensuing value of t_M was obtained as a

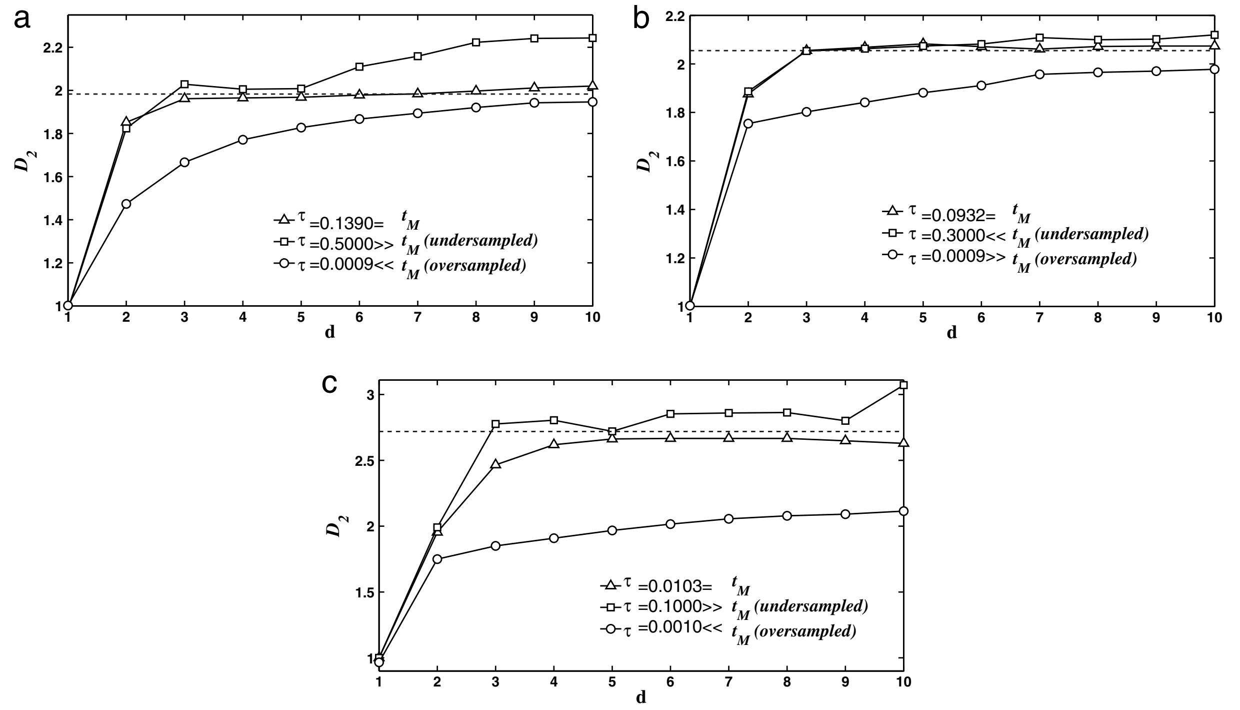

Fig. 7. Mean value over realizations of the correlation dimension ($\langle D_2 \rangle$) for sampled x (y and z samplings produce similar results) as a function of the embedding dimension d: (a) Rossler system with $a = 0.45$, $b = 2$, $c = 4$. (b) Lorenz system with $\sigma = 16$, $r = 45.92$, and $b = 4$. (c) $\mathbf{B}_7$ system with $K = 0.5$, $\alpha = 7.0$, and $\epsilon = 0.23$. In all the cases the time series length considered had $M = 10^5$ *data*. For over-sampled series the value of D_2 remains below the reference value for any embedding dimension. For under-sampled series all linear and nonlinear correlations are lost and the system behaves like a stochastic one. The correlation dimension D_2 increases with the embedding dimension d_e.

function of the signal to noise S/N ratio (σ). In all cases, the value of t_M remained constant (up to three significant decimal digits) for $S/N > 20$ dB. If the noise strength is higher, t_M suddenly decreases as the representative point in the plane $H^{(BP)} \times C^{(BP)}$ moves to the rightwards region (stochastic region), where ordering patterns are strongly affected by noise.

4. Conclusions

We have shown that a particular complexity measure, the SMC-one $C^{(BP)}$, allows one to determine, à la Takens, convenient sampling periods in the case of three paradigmatic non linear models. These sampling periods were called t_M and these models constitute significant examples we feel confident in conjecturing that the optimality criterion used to find t_M may be useful in other scenarios as well, since $C^{(BP)}$ is a measure of the geometric structures produced by nonlinear correlations, always present in this class of dynamical systems.

We have here seen that t_M is compatible with those specific time periods recommended in the literature as adequate delay ones in Takens' reconstruction. Our results closely approach the exact Nyquist–Shannon reconstruction. This is so because high frequencies of the Fourier Spectrum result from chaotic oscillations, and this is certainly the case for the systems studied here. We propose thus that our procedure may be regarded as a methodology that deserves the attention of experimentalists.

To our knowledge, controlled experiments for different sampling times of the measured variables and large numbers of data have not yet been reported in the literature and the highest sampling frequency allowed by the digital acquisition system is the one usually adopted (as an exception see Ref. [21]). The correct t_M-value seems to constitute a clever choosing of the sampling period, that permits one to cover the whole attractor basin and retain in the reconstruction process the most important properties of chaotic systems. In view of the extensive use of digital acquisition systems in all kinds of experiments, we hope that experimentalists may consider the present contribution as a possible useful practical tool in their activities.

Acknowledgments

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